The Application of Mapping in Composition and Design

Paul Doornbusch
Interactive Information Institute, RMIT University
email: pauld@iii.rmit.edu.au

Abstract

Mapping concerns the connection between structures, or gestures and audible results in a musical performance or composition. This is important to computer music composition as most of it is algorithmic in some way or another and it involves mapping. Algorithmic composition is sometimes the process of imagining a gesture or structure - perhaps physical or visual - and then applying a mapping process to turn that structure of the conceptual domain into sound which may display the original conception in some way. This article looks at mapping from the point of view of Australasian algorithmic composition practice, particularly where persistence is an issue, such that the structure (conceptual domain) is embodied and perceptible in the musical result. An attempt is then made to draw some parallel to the practice of (visual) design that uses mapping and examine the similarities and differences with algorithmic music composition.

1 Introduction

Mapping as part of algorithmic composition has not been as thoroughly investigated as mapping in instrument design. This is possibly because it is usually an integrated part of the composition process and not seen as a discrete stage of practice. Mapping in algorithmic composition is different from mapping in instrument design because composition is a process of planning and instruments are for real time music production. The use of mapping which is integral to an instrument has both differences and similarities to the mapping that is an inherent part of algorithmic composition. This paper investigates how some Australasian composers use mapping in their practice and moves towards a taxonomy of the process.

Another aspect of this investigation is how mapping is used by architects and designers. Some designers say that they explicitly use mapping in their practice. There is an attempt to analyse this in a small way and see if there are parallels with composition practice.

2 Defining The Terms

Compositional structures and ideas can take many forms, but they are often abstract in some way, to a greater or lesser degree, from the music that is composed. Composers sometimes use visual ideas of shapes, mathematical functions, physical processes, or phenomena and so on as ideas for creating music. Mapping is the process of taking the (possibly abstract) compositional structures and generating musical parameters. Gesture is an ill-defined term, which for the purposes of this paper means a musical concept; it is not a physical movement.

A musical gesture is a planned change (randomness can be planned) in musical parameters as part of a piece of music. The parameters could be, for example; timbre, density, intensity, timing, pitch, and so on. A compositional gesture is the underlying conception, structure and planning of the musical gesture.

"Conceptual domain" is a term that will be used to cover the entire conceptual area of compositional practice, which includes other organisation strategies as well as compositional gesture.

3 Mapping In Composition

Algorithmic composition can be the practice of composing music through manipulating a number of compositional concepts, structures or gestures to produce a musical piece. The manipulation is usually part of the compositional process, where a multitude of gestures or structures are arranged and then through another process, possibly very direct, musical parameters are extracted. This is most usually done with the aid of a computer, although it should be noted

that this is usually a matter of efficiency and expediency only. It is possible to do this from the micro-composition scale of sound synthesis (where it can only be done with a computer) through to the macro scale of the structural model of the piece. Mapping is the process of extracting the musical parameters from the compositional gestures or structure, such that one set of data is *mapped* onto another. It should also be noted that algorithmic composition might use specific organisations of data that do not necessarily require "gestures"; for example, Markov chains, finite-state machines, fractals and space grammars. These constructs may also require significant mapping processes to output musical data.

There may appear to be some similarities between the methods of mapping in algorithmic composition and the mappings used in the general practice of the sonification of data. This can sometimes be a valid observation with some pieces of music, and there is certainly some overlap between the two areas. However, as the intent of sonification is to illustrate a process or some data, it is primarily instructional. This type of mapping is not the focus of this paper, which is primarily about *composition* and mapping, which is somewhat more ephemeral. The closest sonification comes to composition, what Larry Polansky calls "manifestation" (Polansky 2002), is where the intent is to creatively use a type of sonification of a formal or mathematical process to create a musical idea. For the purposes of this paper, that process will be included as a part of composition.

4 Historical Examples

Mapping has been used by some composers for many years, although it may not have been identified as a separate element of practice. These historical examples will help to explain the function of mapping, showing how it has been used by composers.

A famous example of the mapping process is the part of *Pithoprakta* (1955-56) as reproduced in Formalized Music pp17-21 (Xenakis 1991). Xenakis used the Brownian motion of gas particles, combined with Bernoulli's Law of Large Numbers, as his basic model for the cloud pizzicato-glissandi section. After calculating, statistically, over 1000 velocities of gas particles at given instants of time (the measurement of this was impossible), he then graphed them on an XY plane and directly mapped the straight lines of the velocities to glissandi for 46 string instruments. Xenakis divided his graph vertically into 15 pitched sections, each corresponding to a major third. This was then mapped to the ranges of the string instruments. The mapping was directly of pitch in the vertical direction. This is a particularly direct and concrete example of mapping. All intensities and durations are the same, but to ensure the sensation of a cloud of particles, Xenakis used a complex temporal arrangement of overlapping timing subdivisions that are factorially-unrelated. This is a complex mapping of linear time, designed to represent the instantaneous nature of the movement of the gas particles. Along with many other algorithmic composers, the mapping phase is implicit in much of Xenakis' work. He often used direct mapping as a result of the deliberate organisation of one set of data in such a way that it maps directly to musical parameters.

Another famous example of algorithmic composition and mapping is Charles Dodge's *Earth's Magnetic Field* (1970). Here, Dodge used data from the effects of the radiation of the sun on the magnetic field of Earth. A Bartels diagram showed fluctuations in the Earth's magnetic field for 1961 and this data formed the basis for the piece. Dodge mapped this data, the Kp index (a measure of the average magnetic activity of the earth) to pitches and rhythms. From the program notes of the recording of *Earth's Magnetic Field* (Dodge, 1970), we may glean an insight into the mapping used:

"The succession of notes in the music corresponds to the natural succession of the Kp indices for the year 1961. The musical interpretation consists of setting up a correlation between the level of the Kp reading and the pitch of the note (in a diatonic collection over four octaves), and compressing the 2,920 readings for the year into just over eight minutes of musical time." (Dodge, 1970)

While the pitches appear to be a direct mapping from the Kp index, some elements of the composition such as the timbres were chosen purely for aesthetic effect. An arrangement of the data that plotted the length of sections of the data against the maximum amplitude in the section was used to determine the speed and direction of the sound spatialisation and the rhythms. The data was also sometimes read multiple times to generate the musical parameters. That, combined with the similarity of the fluctuations in the Kp index to 1/f noise data, contributes to the aspects of self-similarity in the piece.

The two previous examples contrast different approaches to mapping. It may be linear and direct, but it may also be nonlinear and more complex. Both examples use the data as a structural component and the music achieves some structural unity for that. Also, both of these examples come from a time when computer music practice was in its infancy. With current tools and thinking, composers can use mapping in much more creative and indirect ways.

5 How Mapping Is Used

There has not been the same formalisation of algorithmic composition and mapping as there has been for other musical composition. Additionally, there is little formal analysis of music in terms of mapping or compositional gesture. As there is no set method for defining the mappings or structures, each composer tends to use their own methods for their own reasons.

Compositional structures and mappings are also used differently by different composers, thus making this a problematic area for analysis and study: For example, the structural model and mapping for a piece of music will have different applications for a composer who is focused on, say, spectral composition in contrast to another focused on some other form of algorithmic composition.

It is clear from descriptions of how composers use mapping between the conceptual or abstract domain and the musical domain, that it is sometimes used in a similar combination of ways to how it is used in instrument design. That is; mapping one compositional parameter to many musical parameters (one to many), mapping many compositional parameters to one musical parameter (many to one) and mapping many compositional parameters to many musical parameters (many to many). Additionally, there is another parallel between instrument designers and composers, the mappings themselves may be linear mappings or nonlinear mappings (Hunt and Wanderley 2000, Hunt, Wanderley and Kirk 2000). However, in compositional mapping there is the additional possibility of repetitive nonlinear mappings and most importantly, each composer has their own combination of mapping techniques.

From the comments made by the architectural designers questioned, it seems that there is some overlap in the use of mapping to move from the abstract to the concrete. This is an area that is only touched upon here, to demonstrate that this is (yet another) intersection between music and architectural design.

6 Questions Of Mapping

To understand how Australasian composers use mapping, a number of them were asked questions concerning their practice. The broad answers are summarised below. The composers were; Roger Alsop, Rodney Berry, Chris Cree Brown, Phil Brownlee, Warren Burt, Densil Cabrera, Tim Kreger, Peter McIlwain, Gordon Monroe, Garth Paine, Greg Schiemer, and Paul Doornbusch. The five designers questioned were Pia Ednie-Brown, Tom Kovak, Paul Minifie, Vivian Mitsogianni and Julian Raxworthy. For the designers the questions were modified to exchange “architectural design” for “composition”, “viewer” for “listener” and “visual output” for “musical output”, and so on. The questions, asked were:

1. Is mapping something that you are conscious of when you are composing?

2. Do you have a consistent approach to mapping and (algorithmic) composition or does it vary and why?

3. When implementing a mapping strategy for (part of) a composition, do you organise this in a particular ‘analytical’ way (decomposing the problem in a technical manner), or in a more creative and holistic way for a purely aesthetic result?

4. Is the mapping component of your compositions something that might be perceptible by a listener, or of interest to them and why?

5. Is the mapping component of algorithmic composition something that is pre-determined for you or is it part of a process of exploration?

6. Do you use individual mapping strategies for individual parameters or is there reuse of mapping strategies or a global system? (ie are they monoparametric or multiparametric?)

7. What elements do you control algorithmically in a composition and can you comment on the function and importance of these?

8. Is the mapping consistent within these elements or not?

9. Are the mapping schemas you use mostly linear mappings or non-linear in some fashion, and why?

10. Can mapping be considered a composition technique in itself?

There was deliberate overlap in the questions to elicit as much information as possible and the respondents were also asked to make a yes/partly/no, multiple-choice response to each question.

7 Responses

The results of the composers' responses are summarised below, making it possible to note similarities and divergences. The answers supplied by the composers will be provided in an appendix to another, larger, paper that can be downloaded or will be sent to anyone requesting it. In this summary, it is possible to achieve the an initiall overview of this area.

7.1 General remarks by respondents:

While composers may use mapping to move from the conceptual domain to musical parameters, mapping can work backwards for listeners, providing access to structures and concepts from the music. Composers tend to use complex ideas and structures for music. Focusing on mapping assumes that something pre-exists before the act of composition and not all composers may think or work like that. If mapping is a means of moving from the conceptual domain to the concrete domain, then it may be important for all composition, but it is central to algorithmic composition.

7.2 Collated Responses:

The tables below show the general response versus the number of composers (C) or designers (D) who had that response. While the number of people questioned means that this is a statistically minuscule sample, it does point to trends and variations.

Question 1.	C	D
General Response	#	#
Yes, I am conscious of using mapping	8	3
Sometimes I am conscious of using mapping	3	2
No, I never consciously use mapping	0	0

Question 2.	C	D
General Response	#	#
I have a consistent approach to mapping	0	0
I sometimes have a consistent approach to mapping	5	3
I do not have a consistent approach to mapping	6	2

Question 3.	C	D
General Response	#	#
I mostly use mapping analytically	0	1
Mapping is sometimes used analytically	3	2
I mostly use mapping creatively	6	2

Question 4.	C	D
General Response	#	#
Mapping is of great interesting for listeners	2	1
Sometimes mapping is interesting to listeners	9	4
Mapping is not interesting to listeners	0	0

Question 5.	C	D
General Response	#	#
Mapping is predetermined in composition	1	0
Mapping is sometimes predetermined, sometimes part of exploration	4	2
Mapping is part of the exploration of composition	6	3

Question 6.	C	D
General Response	#	#
I use individual mapping strategies for individual parameters (monoparametric)	2	0
Both individual mappings strategies are used as well as reusing some	7	2
I reuse mapping strategies for multiple parameters (multiparametric)	1	3

Question 7.	C	D
General Response	#	#
I usually control as many elements as possible	2	1
I control most elements sometimes, fewer at other times	4	0
I usually control some elements only	5	2

Question 8.	C	D
General Response	#	#
The mapping is mostly consistent for each element	9	3
The mapping is sometimes consistent for each element	1	0
The mapping is mostly not consistent for each element	1	2

Question 9.	C	D
General Response	#	#
Mostly linear mappings are used (very complex structures)	1	1
A mixture of linear and nonlinear mappings are used (depends on structure)	7	1
Mostly nonlinear mappings are used (simpler structures)	3	3

Question 10.	C	D
General Response	#	#
Mapping can be a compositional technique in itself	6	3
Mapping is part of compositional technique	5	2
Mapping cannot be considered a compositional technique at all	0	0

8 Types Of Mapping

From the composers' descriptions of how they use mapping, it seems that there is no set method for mapping data from the domain of the conceptual, gestural or structural to the musical domain. Linear and simple mappings are *sometimes* used by composers, but it is used significantly less often than mappings that are more complex.

Simple mappings have been most clearly described as ratiometric (Polansky 2002). In this type of mapping, a doubling of the data to be mapped results in a "doubling" of the musical parameter. This could be, for example, an octave pitch displacement or a doubling of the loudness. Note that this second linear mapping might be perceptually linear to a particular person under particular conditions, but it would be mathematically nonlinear. Because ratiometric mappings are the easiest to perceive they are particularly useful when the composer has some data, gesture or concept that should be translated as directly as possible for the listener. The previous example of the pitch mappings from Xenakis' Pithoprakta is an illustration of this. This particular example is musically successful because the data to be mapped is very complex from the outset. Simple mappings can be less musically successful if the data to be mapped is very simple.

Complex mappings may be regularly nonlinear, such as an exponential law. For example, a square law will produce a fourfold increase in the musical parameter from a doubling of the data to be mapped. If this is perceptually based it can be perceived as a more extreme or less extreme mapping with changes in the data to be mapped. Other types of complex mappings are as varied as can be imagined. They can sometimes be related to a complex arrangement of the data to be mapped, or other potentially chaotic functions can be involved. This may obscure the original concept, compositional gesture, or data, or it may embellish it and give it another dimension. However, there is clearly a limit to how far such complexity in mapping can be taken before all sense of the original data is lost in the mapping. Polansky convincingly suggests that the cognitive weight of complex mappings degenerates rapidly and nonlinearly such that beyond a certain point everything is just "complex".

It has been a repeated outcome from instrument design research that humans prefer complex mappings (Hunt and Kirk 2000). This would appear to translate somewhat into the domain of composition, given the previous situation that linear mappings may be most appropriate for very complex data.

In one way, the variety of sophisticated, creative and exploratory approaches to mapping, as embodied in the works of algorithmic composers, is a wonderful outcome and it ensures the variety of music from algorithmic composition. In another way, it means that mapping strategies are reinvented by every composer and young composers do not have easily accessible models to work from. If there was created a categorisation of compositional mapping strategies and a catalogue was built of mapping techniques, then other composers could refer to this and build on it in their own practice. This would be of particular use to student composers (Polansky 2002). In this way, mapping in algorithmic composition could be demystified and more complex, varied and musically appropriate practices could be developed by building on the work of others. The counter argument to this is that there is something valuable in the effort of developing sophisticated mappings for oneself. In instrument design there is certainly interest in the effort required to play an instrument and how mapping relates to this (Ryan 1992).

The answers from the designers are interesting for a number of reasons. There appears to be some degree of overlap in the way that composers use mapping and how architectural designers use it. This is an additional parallel than those that are usually drawn between architecture and music; that they both communicate through, and are deeply concerned with, the concept of form, and that one divides space while the other divides time. The similarities in the ways both designers and composers use mapping, and also how they do not use mapping (the 0 scores in the tables above), points to some similarity in the creative process between the disciplines. However, I think there is both too much semantic variation in the understanding of the questions presented, and too few

people involved, to make anything more than a general claim of similarity.

Something which has not been explored here, but which would be of interest to some composers, are cultural associations with mapping. For example, are there culturally invariant mappings with some parameters such as pitch or intensity with height? One example from instrument design that has caused problems is the original Theremin mapping of intensity with proximity to a horizontal antenna such that to make a louder sound the performer would bring their hand *down* to the volume antenna. This was done for practical reasons so that without a performer there is no sound, but modern theremins allow this to be swapped because of the cognitive dissonance caused by the playing action.

9 Conclusions

One clear outcome of this is that the questions need improving if this is going to be attempted again. This is indicated clearly in the detailed responses where not all of the questions seemed sensible to some people, but it is also evident in the spread of answers to some of the questions, although some spread is to be expected of course. In addition, the semantic understanding of many of the terms surrounding mapping and composition (and design) seems to be ill defined and this leads to a varied understanding of the questions and therefore variable answers.

Not withstanding the problems with the questions and semantics, some valuable information can be extracted from the answers. The answers indicate that composers who use algorithmic techniques certainly use mapping from one domain to another, often in very sophisticated ways. Also indicated is that there is no ideal solution or single method, but that it is largely a process of exploration. Very simple or obvious mappings are sometimes problematic in composition probably because of the oversimplification of the resultant music, but they may be appropriate under some circumstances such as with very complex structures or data. More complex mapping strategies are more common and more musically useful, but very numerous and their application and musical usefulness appears to be an aesthetic judgement of the composer. Composers appear to be aware of complexity that does not scale linearly for listeners such that there is some limit to what makes sense to a listener's perceptions and beyond this it is so complex that there is no sense of the original concept or data. There seems to be almost as many approaches to mapping as there are composers or designers. It is the character of composition, being an expansive and creative activity, that if ever a theory provided a "standard", "optimised" or "ideal" mapping technique or repertoire, that composers would ignore it and go beyond that. Having said that, there would appear to be *something* to gain by creating a catalogue of historical and current mapping practices, even if only in a pedagogical sense (Polansky 2002) and this could be started by expanding on the questions presented above.

The nature of composition and composers means that there will never be "solution" to the mapping problem in algorithmic composition, that it will remain a part of the technique and exploration of composers.

Not being a designer, I am unqualified to comment on the area of design, but the survey showed significant areas of similarity in the responses. Significant enough, I feel, that there is some similarity of practice and theoretical approach such that further investigation and cross-discipline discourse is warranted to inform both disciplines.

10 Acknowledgments

I would like to thank and gratefully acknowledge the contributions of the composers and designers who made much of this work possible. The composers are; Roger Alsop, Rodney Berry, Chris Cree Brown, Phil Brownlee, Warren Burt, Densil Cabrera, Tim Kreger, Peter McIlwain, Gordon Monroe, Garth Paine, Greg Schiemer. The designers are Pia Ednie-Brown, Tom Kovak, Paul Minifie, Vivian Mitsogianni and Julian Raxworthy.

Also, much of the material in this paper is the result of stimulating conversations with Paul Berg, Larry Polansky and Marcelo Wanderly.

References

Wanderley, M. M. and Battier, M. eds. 2000. *Trends in Gestural Control of Music*. Paris: IRCAM

Cadoz, C. and Wanderley, M. M. 1999. *Gesture-Music*. In Trends in Gestural Control of Music. Paris: IRCAM CDROM.

Dodge, C. 1970. *Earth's Magnetic Field program notes*. In Nonesuch LP H 71250 EARTH'S MAGNETIC FIELD: Realizations in Electronic Sound.

Dodge, C. and Jerse, T. 1997. *Computer Music second edition*. New York: Schirmer Books

Hunt, A., Wanderley, M. M. and Kirk, R. 2000. *Towards a Model for Instrumental Mapping in Expert Musical Interaction*. In Proceedings of the 2000 International Computer Music Conference. San Francisco: International Computer Music Association, pp. 209 - 212.

Hunt, A. and Wanderley, M. M. (eds.) 2000. *Mapping of Control Variables to Musical Variables*. Interactive Systems and Instrument Design in Music Working Group. Website: http://www.notam02.no/icma/interactivesystems/mapping.html

Hunt, A. and Kirk, R. 2000. *Trends in Gestural Control of Music*. Paris: IRCAM

Kramer, G. 1996. *Mapping a single data stream to multiple auditory variables: A subjective approach to creating a compelling design*. In Proceedings of International Conference on Auditory Display - ICAD'96.

Matossian, N. 1986. *Xenakis*. London: Kahn & Averill

Polansky, L. 2002. *Manifestation and Sonification*. Unpublished paper.

Roads, C. 1996. *The Computer Music Tutorial*. Cambridge, Massachusettes: The MIT Press

Ryan, J. 1992. *Effort and Expression*. In Proceedings of the 1992 International Computer Music Conference. San Francisco: International Computer Music Association, pp. 414 - 416.

Thieberger, E. M. 1995. *An Interview with Charles Dodge*. Computer Music Journal 19(1): 11-25

Varga, B. A. 1996. *Conversations with Iannis Xenakis*. London: Faber and Faber

Xenakis, I. 1991. *Formalized Music Pendragon Revised Edition*. New York: Pendragon Press

Appendix 1: Detailed Responses

This is a one page version of the detailed responses to the first question. The rest of the responses are on the ACMC 2002 website where there is a downloadable version of the whole paper. See http://www.iii.rmit..edu.au/sonology/ACMC2002

The various approaches to mapping in algorithmic composition, and design, can be easily seen in the responses. Additionally, the comments by the designers show the overlap and differences with composers in thinking about mapping. The composers questioned were; Roger Alsop (**RA**), Rodney Berry (**RB**), Chris Cree Brown (**CCB**), Phil Brownlee (**PB**), Warren Burt (**WB**), Densil Cabrera (**DC**), Tim Kreger (**TK**), Peter McIlwain (**PM**), Gordon Monroe (**GM**), Garth Paine (**GP**), Greg Schiemer (**GS**), and Paul Doornbusch (**PD**). The designers questioned were Pia Ednie-Brown (**PEB**), Tom Kovak (**TXK**), Paul Minifie (**PXM**), Vivian Mitsogianni (**VM**) and Julian Raxworthy (**JR**).

1. Is mapping something that you are conscious of when you are composing (designing)?

PM: In the sense that mapping is some process that transforms material from state to another then all standard compositional procedures (such as transposition, inversion etc) are a form of mapping. I am not always conscious of these procedures as they have become more automatic over time. However, the term mapping is often used in a specific way to refer to transformations of musical materials in correlation with extra-music materials. When I have done this I am usually very aware of the process chiefly because I feel that must make judgments about the effectiveness in musical terms (than is, my own subjective musical definitions) of this, or that, mapping procedure.

WB: If the piece I'm doing involves the application of some sort of numerical data, whether a simple use of a random number generator, or something more complex, I'm concerned with mapping. Numerical sources only assume meaning when they're applied to materials that have a kind of "graspability" to them. That is, to hear a number pattern for its own sake, it might be better at first to apply it to a set of 5 pitches rather than all 12 (if we're talking 12tone equal temperament for the moment), because with fewer pitches we might be able to "forget" the traditional musical implications of the fewer pitches (ie, they're so bland we can forget them for the moment) and concentrate on the qualities of our numerical structure. When it comes time to put those structures into a larger "piece", then a much more conscious selection of materials can take place.

GM: Most, but not all, of my compositions involve mapping in some way.

DC: It is not a term that I normally apply. However, I'm conscious of the ideas of 'transcription' or 'sonification', which seem to be closely related. In some cases, my works have realised the acoustic implications of a spatial journey, and I suppose the term 'mapping' is especially apt there.

GP: When composing a piece of music, or a soundscape, I view the mapping of the sound material to the intended emotional context of the work to be central to the craft of composition. I am usually

contemplating at least two layers of material at any one time - an environment/broader context, and individual lines of activity/dialogue/narrative.

PD: Yes, I'm conscious of mapping when I'm working, but it's not the first thing I think of. I'm very conscious of it when I'm working on transcribing a structure or gesture into music or sound. However, I am not sure if I think that it is important, in my work, that the compositional structures or gestures are somehow transmitted to the audience. In a way it is, but mostly I'm trying to give the audience a powerful musical experience, so as far as transmitting a gesture or structure is important for that, then yes it is important. I try to map from compositional gestures and structures to musical parameters in such a way that the gesture or structure is evident when *I* listen to the result, but this may not translate to these being perceived by the audience. Having said that, most of the compositional gestures and structures I use are abstract concepts and I usually have movement in three or four aspects of them simultaneously, otherwise I find the result boring musically.

PEB: Mapping is always an integral part of architectural design. All aspects of design involve representational devices The act of moving between various modes of representation involves many types or modes of mapping. As such, mapping is integral and inevitable; some aspects of which are more conscious than others.

The question gathers a more elaborate and more precise answer for me when I think of the sculptural work I've done. In this case the outcome is the object itself, there is not an imagined future event of actualisation (building), making what I see as the primary nature of mapping very directly accessible. I made a series of casts that emerged through experiments with latex and plaster. Each one of the series is the product of a different deformation of the same mould. They were made through pouring liquid plaster into the latex mould of a shower tap. The act of making these involved a kind of puppeteering act in which the stretchy, elastic moulds were suspended with string, positioned with wire and held down with tape under chairs, off coat hangers - whatever was at hand. The latex stretched and deformed with the weight of the fluid plaster, particularly as the makeshift supports fell away or shifted. Each casting was enacted with a different placement or 'context' of the mould. Although the process could be guided toward expected results, the final cast was always a surprise, being very contingent to the moment. The resulting series of different casts arose from the coalescing involutions of a myriad of variable relations (ie the outcome is a mapping of the complexity that is involved in the situation - it is a qualitative mapping of an event) all struck into an accord: form extracted from dynamic interactions, falling into one another in a collaborative agreement with a responsively overarching skin. In this example, it is a *situation* that is explicitly mapped more than anything else. I would argue that this is *always* the case in a creative act, no matter what the process, even if the dynamic variability of the moment or situation is embedded (and rendered opaque) within a rigid, linear, direct mapping process. Often, where extremely non-linear mapping positions are taken, there is an assumption that the material being mapped is the subjectivity of the author. What I like about the cast example is that, while being manifestly non-linear, like many programmed acts of mapping, there is an implied shift in the role of the author/creator in that the outcomes are, to a relatively explicit extent, a product of the process (where designing the process itself becomes the primary role of the composer).

PXM: In some projects mapping is of crucial interest - it is a key way of capturing relationships and structures into a design work. It helps because it is an antitectonic strategy, disrupting a normal tectonic set of relations.

VM: My own work deals with a 'machine based' design methodology, that is the process is designed rather than the object as a starting point. The 'process' or 'machine' will often include mapping, diagramming in part. This is a generalisation, but the work generally involves the use of techniques that contain abstract information (such as diagrams, scientific systems of behaviour) and are used as part of the larger design process. This is more akin to a 'machine' based design process in the sense of Deleuze's use of the term 'machine' in that the 'mapping' is a part of the machine that (along with other ideas and techniques) are 'applied' to something, they are not the thing itself. That is, mapping is not the main tool used.